Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In the laboratory, manganese (II) salt is oxidised to permanganate ion in aqueous solution by

(A) hydrogen peroxide

(B) conc. nitric acid

(C) peroxy disulphate

(D) dichromate

Q.2

When Cu reacts with $\mathrm{AgNO_3}$ solution, the reaction takes place is

(A) oxidation of Cu

(B) reduction of Cu

(C) oxidation of $\mathrm{Ag^+}$

(D) reduction of $\mathrm{NO_3^-}$

Q.3

$\mathrm{CuSO_4}$ reacts with excees KCN solution to form:

(A) $\mathrm{Cu(CN)}$

(B) $\mathrm{Cu(CN)_2}$

(C) $\mathrm{K_3\,[Cu(CN)_4]}$

(D) $\mathrm{K_4\,[Cu(CN)_6]}$

Q.4

Which of the following compounds gives red precipitate with $\mathrm{AgNO_3}$?

(A) KI

(B) $\mathrm{K_2CrO_4}$

(C) NaBr

(D) $\mathrm{NaNO_3}$

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Q.5

When horn silver ore is dissolved in excess of sodium cyanide solution compound formed is

(A) AgCN

(B) $\mathrm{Na}\left[\mathrm{Ag(CN)_2}\right]$

(C) $\mathrm{Na_2}\left[\mathrm{Ag(CN)_3}\right]$

(D) $\mathrm{Na_4}\left[\mathrm{Ag(CN)_5}\right]$

Q.6

In the equation $4\mathrm{M} + 8\mathrm{CN^-} + 2\mathrm{H_2O} + \mathrm{O_2} \rightarrow 4\left[\mathrm{M(CN)_2}\right]^- + 4\mathrm{OH}$
Identify the metal M. It is

(A) copper

(B) iron

(C) gold

(D) zinc

Q.7

If $\mathrm{KMnO_4}$ is reduced by oxalic acid in an acidic medium then oxidation number of Mn changes from

(A) 4 to 2

(B) 6 to 4

(C) $+7$ to $+2$

(D) 7 to 4

Q.8

In the preparation of $\mathrm{KMnO_4}$ pyrolusite $\left(\mathrm{MnO_2}\right)$ is first converted to potassium manganate $\left(\mathrm{K_2MnO_4}\right)$. In this conversion, the oxidation state of manganese changes from

(A) $+1$ to $+3$

(B) $+2$ to $+4$

(C) $+3$ to $+5$

(D) $+4$ to $+6$

Q.9

In an alkaline condition $\mathrm{KMnO_4}$ reacts as follows:

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$2KMnO_4 + 2KOH \rightarrow 2\ K_2MnO_4 + H_2O + O$

Its equivalent weight is

(A) 31.6

(B) 52.7

(C) 49

(D) 158

Q.10

In which of the following compounds manganese has oxidation number equal to that of iodine in KIO_4?

(A) Potassium manganate

(B) Potassium permanganate

(C) Manganous chloride

(D) Manganese chloride

Q.11

The colour of $CuCr_2O_7$ solution in water is green because

(A) $Cr_2O_7^{2-}$ ions are green.

(B) Cu^{2+} ions are green.

(C) Both $Cr_2O_7^{2-}$ and Cu^{2+} ions are green.

(D) Cu^{2+} ions are blue and $Cr_2O_7^{2-}$ ions are yellow.

Q.12

A compound of iron exists as a dimer in vapour state. It is hygroscopic in nature and dissolves in water giving brown acidic solution. The compound is

(A) Fe_3O_4

(B) $FeSO_4$

(C) $FeCl_3$

(D) FeO_2

Q.13

$(X) + K_2CO_3 + \text{Air} \xrightarrow{\text{heat}} (Y)$

$(Y) + Cl_2 \longrightarrow (Z)\ \text{Pink}$

Which of the following is correct?

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(A) $X =$ black, MnO_2, $Y =$ Blue, K_2CrO_4, $Z = KMnO_4$

(B) $X =$ green, Cr_2O_3, $Y =$ Yellow, K_2CrO_4, $Z = K_2Cr_2O_7$

(C) $X =$ black, MnO_2, $Y =$ green, K_2MnO_4, $Z = KMnO_4$

(D) $X =$ black, Bi_2O_3, $Y =$ colourless $KBiO_2$, $Z = KBiO_3$

Q.14

$CuSO_4$ (aq.) $\xrightarrow{H_2S\uparrow} M\downarrow \xrightarrow[\text{of KCN}]{\text{Excess}} N + O$

Then final products N and O are respectively.

(A) $[Cu(CN)_4]^{3-}$, $(CN)_2$

(B) $CuCN$, $(CN)_2$

(C) $[Cu(CN)_4]^{2-}$, $(CN)_2$

(D) $Cu(CN)_2$, K_2S

Q.15

Which of the following statements is false ?

(A) $Na_2Cr_2O_7$ is less soluble than $K_2Cr_2O_7$.

(B) $Na_2Cr_2O_4$ is a secondary standard solution in volumetry.

(C) CrO_4^{2-} is tetrahedral in shape.

(D) $Cr_2O_7{}^{2-}$ has a $Cr-O-Cr$ bond.

Q.16

A red solid is insoluble in water. However it becomes soluble if some KI is added to water. Heating the red solid in a test tube results in liberation of some violet coloured fumes and droplets of a metal appear on the cooler parts of the test tube. The red solid is

(A) HgI_2

(B) HgO

(C) Pb_3O_4

(D) $(NH_4)_2Cr_2O_7$

Q.17

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Of Cr (VI) as $Cr_2O_7^{2-}$ and CrO_4^{2-}, which is better oxidising agent?

(A) CrO_4^{2-}, basic medium

(B) $Cr_2O_7^{2-}$, basic medium

(C) $Cr_2O_7^{2-}$, acidic medium

(D) CrO_3, basic medium

Q.18

A blue colouration is not obtained when

(A) ammonium hydroxide dissolves in copper sulphate.

(B) copper sulphate solution reacts with $K_4\,[Fe(CN)_6]$

(C) ferric chloride reacts with sod. ferrocyanide.

(D) anhydrous $CuSO_4$ is dissolved in water.

Q.19

Identify the incorrect statement

(A) $CuSO_4$ reacts with KCl in aqueous solution to give Cu_2Cl_2

(B) $CuSO_4$ reacts with KI in aqueous solution to give Cu_2I_2

(C) $CuSO_4$ react with $NaOH$ and glucose in aqueous medium to give Cu_2O.

(D) $CuSO_4$ on strong heating gives CuO.

Q.20

The black compound formed during the reaction between sodium thiosulphate and silver nitrate is

(A) silver thiosulphate $(Ag_2\ S_2O_3)$

(B) silver sulphide $(Ag_2\ S)$

(C) silver sulphate (Ag_2SO_4)

(D) silver sulphite (Ag_2SO_3)

Answers & Solutions

Q.1 Answer:

peroxy disulphate

Solution:

In laboratory, manganese (II) ion salt is oxidised to permagnate ion in aqueous solution by peroxodi-sulphate.

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$$2Mn^{2+} + S_2O_8^{2-} + 8H_2O \longrightarrow$$

peroxodisulphate ion

$$2MnO_4^- + 10SO_4^{2-} + 16H^+$$

Q.2 Answer:

oxidation of Cu

Solution:

$Cu^0 + 2AgNO_3 \rightarrow Cu^{2+}(NO_3)_2 + 2Ag$

It is oxidation of copper $\left(Cu^0 \rightarrow Cu^{2+}\right)$

Q.3 Answer:

$K_3[Cu(CN)_4]$

Solution:

$2CuSO_4 + 10KCN \longrightarrow$

$2\ K_3[Cu(CN)_4] + 2\ K_2SO_4 + (CN)_2$

Q.4 Answer:

K_2CrO_4

Solution:

$2AgNO_3 + K_2CrO_4 \rightarrow$

$Ag_2CrO_4 + 2KNO_3$

Red precpitate

Q.5 Answer:

$Na[Ag(CN)_2]$

Solution:

$AgCl + 2NaCN \rightarrow$

$Na[Ag(CN)_2] + NaCl$

Q.6 Answer:

gold

Solution:

This is cyanide process for gold $(M = Au)$.

Q.7 Answer:

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$+7$ to $+2$

Solution:

In acid medium

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

(O.S. of Mn changes form $+7$ to $+2$).

Q.8 Answer:

$+4$ to $+6$

Solution:

In MnO_2 (O.S. of Mn is $+4$) in K_2MnO_4 (O.S.of Mn is +6). Hence O.S. changes by 2 .

Q.9 Answer:

158

Solution:

Change in O.S. is by 1. Hence eq. wt. is $\frac{158}{1} = 158$

Q.10 Answer:

Potassium permanganate

Solution:

In KIO_4, O.S of I is $+7$ and in $KMnO_4$, O.S. of $Mn$is $+7$

Q.11 Answer:

Cu^{2+} ions are blue and $Cr_2O_7^{2-}$ ions are yellow.

Solution:

Green colour of $CuCr_2O_7$ is due to blue colour of Cu^{2+} ions and yellow colour of $Cr_2O_7^{2-}$ ions

Q.12 Answer:

$FeCl_3$

Solution:

The solution is acidic due to formation of HCl and brown due to $Fe(OH)_3$

$FeCl_3 + 3H_2O \rightarrow Fe(OH)_3 + 3HCl$

Q.13 Answer:

$X =$ black, MnO_2, $Y =$ green,
K_2MnO_4, $Z = KMnO_4$

Solution:

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$$2MnO_2 + 2K_2CO_3 + O_2 \xrightarrow{\Delta}$$
(X) (air)

$$2K_2MnO_4 + 2CO_2 \uparrow$$
green
(Y)

$$2K_2MnO_4 + Cl_2 \longrightarrow 2KMnO_4 + 2KCl$$
(Y) (Z) Pink

Q.14 Answer:

$[Cu(CN)_4]^{3-}$, $(CN)_2$

Solution:

$$CuSO_4(aq.) \xrightarrow{H_2S} \underset{\text{Black ppt.}}{2CuS} \xrightarrow[\text{Excess}]{KCN}$$
(M)

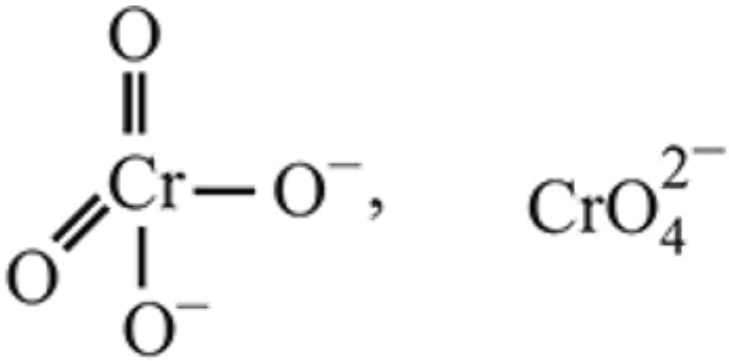
$$[Cu(CN)_4]^{3-} + (CN)_2 \uparrow$$
(N) (O)

Q.15 Answer:

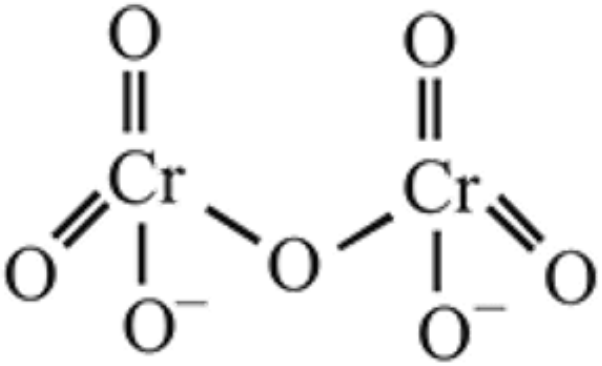
$Na_2Cr_2O_7$ is less soluble than $K_2Cr_2O_7$.

Solution:

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CrO_4^{2-} is tetrahedral



, Cr – O – Cr bond in $Cr_2O_7^{2-}$

$Na_2Cr_2O_4$ **is a secondary standard solution in volumetry as it is hygroscopic in nature. Hydration energy of** Na^+ **is greater than** K^+ **because of smaller size of** Na^+**. So,** $Na_2Cr_2O_7$ **is more soluble than** $K_2Cr_2O_7$

Q.16 Answer:

HgI_2

Solution:

When KI **is added to mercuric iodide it disssolve in it and form complex.**

$$\underset{\text{red, solid (insoluble)}}{HgI_2} + KI \rightarrow \underset{\text{(soluble)}}{K_2[HgI_4]}$$

On heating HgI_2 **decomposes as** $HgI_2 \rightleftharpoons Hg + I_2$

(violet vapours)

Q.17 Answer:

$Cr_2O_7^{2-}$, acidic medium

Solution:

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$CrO_4^{2-}(aq) + 4H_2O + 3e^- \longrightarrow$
$Cr(OH)_3 + 5OH^-,$
$E^o = -0.13\ V$
$Cr_2O_7^{2-}$ **(aq.)** $+14H^+$ **(aq.)** $+6e^- \longrightarrow$
$2Cr^{3+}$ **(aq.)** $+7H_2O,$
$E^o = 1.33\ V$

Q.18 Answer:

copper sulphate solution reacts with
$K_4\,[Fe(CN)_6]$
Solution:

$2CuSO_4 + K_4\,[Fe(CN)_6] \longrightarrow$
$Cu_2\,[Fe(CN)_6] + 2\,K_2SO_4$
Chocolate ppt.

Q.19 Answer:

$CuSO_4$ reacts with KCl in aqueous solution to give Cu_2Cl_2
Solution:

$2CuSO_4 + 4KI \rightarrow Cu_2I_2 + 2\,K_2SO_4 + I_2$
(not given by KCl). Fehling solution gives Cu_2O with reducing substances

$$CuSO_4 \xrightarrow[\Delta]{\text{strong}} CuO + SO_3$$

Q.20 Answer:

silver thiosulphate $(Ag_2\,S_2O_3)$
Solution:

$$2AgNO_3 + Na_2S_2O_3 \rightarrow$$
$$Ag_2S_2O_3 + 2NaNO_3$$

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